

INDU 211 · Worked Problems & Quantitative Analysis

Cost-Volume & Break-Even Engineering Decision Guide

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1. The Mathematical Framework of Cost-Volume Analysis

In production systems, engineering decisions are justified by numbers. Every production engineer must master two quantitative models:

1. **Single-Process Revenue Break-Even Analysis** (Evaluating whether a product is economically viable against market price).
2. **Multi-Process Crossover Selection** (Choosing the most cost-effective manufacturing process at a given production volume).

Fundamental Equations

$$\text{Total Cost: } TC(Q) = FC + v \cdot Q = b + a \cdot Q$$

$$\text{Total Revenue: } TR(Q) = P \cdot Q$$

$$\text{Total Profit: } \Pi(Q) = TR(Q) - TC(Q) = (P - v)Q - FC$$

$$\text{Break-Even Quantity: } Q_{BEP} = \frac{FC}{P - v} = \frac{b}{P - a}$$

$$\text{Two-Process Crossover Volume: } Q_{1,2} = \frac{FC_1 - FC_2}{v_2 - v_1}$$

Where:

- Q : Annual production volume (units/year).
- FC (or b): Fixed annual capital cost (\$).
- v (or a): Variable direct production cost per unit (\$/unit).
- P : Selling price per unit (\$/unit).
- $(P - v)$: Contribution margin per unit (\$/unit).

2. Problem 1: Single-Process Viability (Lecture Slide 12)

Problem Statement

A manufacturing enterprise is considering producing a new mechanical component with the following financial parameters:

- **Annual Fixed Cost (FC):** \$28,000 (tooling, machine rental, space).
- **Variable Cost per Unit (v):** \$100/unit (raw material, direct machining labor).
- **Selling Price (P):** \$200/unit.

Questions

1. Determine the annual Break-Even sales volume Q_{BEP} .
2. If projected market demand is $Q = 220$ units/year, what is the annual net profit or loss? Should the system be installed?
3. If market demand is $Q = 500$ units/year, calculate total profit.
4. What minimum selling price P_{min} allows the company to break even at only $Q = 200$ units?

Step-by-Step Solution

Part 1: Break-Even Volume

At break-even, total revenue equals total cost:

$$TR(Q) = TC(Q) \implies 200Q = 100Q + 28,000$$

$$100Q = 28,000 \implies Q_{BEP} = \frac{28,000}{100} = \mathbf{280 \text{ units}}$$

Part 2: Profit at $Q = 220$ units

$$\Pi(220) = TR(220) - TC(220) = (200)(220) - [28,000 + 100(220)]$$

$$\Pi(220) = 44,000 - [28,000 + 22,000] = 44,000 - 50,000 = \mathbf{-\$6,000 \text{ (Loss!)}}$$

Managerial Conclusion: The company must **reject** installation. Selling 220 units fails to recover fixed overhead, resulting in an annual loss of \$6,000.

Part 3: Profit at $Q = 500$ units

$$\Pi(500) = (P - v)Q - FC = (200 - 100)(500) - 28,000 = 50,000 - 28,000 = \mathbf{+\$22,000 \text{ (Net Profit)}}$$

Part 4: Target Selling Price for $Q = 200$ units

$$P \cdot (200) = 28,000 + 100(200) \implies 200P = 28,000 + 20,000 = 48,000$$

$$P_{min} = \frac{48,000}{200} = \$240/\text{unit}$$

3. Problem 2: 3-Process Selection Problem (Lecture Slide 15)

Problem Statement

A manufacturing firm can produce a stamped structural bracket using three alternative production methods:

Process Option	Description	Fixed Annual Cost (FC)	Variable Cost (v)
Process A	High-speed automated stamping press	\$110,000	\$2.00/unit
Process B	Semi-automated CNC turret punch	\$80,000	\$4.00/unit
Process C	Manual brake press & notch tooling	\$75,000	\$5.00/unit

Question A: Annual demand is projected at $Q = 10,000$ units/year. Which process should be selected?

Step 1: Calculate Total Cost for each alternative at $Q = 10,000$ units

$$TC_A(10,000) = 110,000 + (2.00)(10,000) = 110,000 + 20,000 = \$130,000$$

$$TC_B(10,000) = 80,000 + (4.00)(10,000) = 80,000 + 40,000 = \mathbf{\$120,000} \quad (\text{Minimum Cost!})$$

$$TC_C(10,000) = 75,000 + (5.00)(10,000) = 75,000 + 50,000 = \$125,000$$

Step 2: Compare and Conclude

$$\mathbf{TC_B(\$120,000) < TC_C(\$125,000) < TC_A(\$130,000)}$$

Optimal Decision at 10,000 units: Select **Process B**. It saves \$5,000 compared to Process C and \$10,000 compared to Process A.

Question B: At what production volume is each process preferred?

To determine the exact volume thresholds, we calculate the pairwise crossover points:

Crossover 1: Process B vs. Process C

Equating $TC_B(Q) = TC_C(Q)$:

$$80,000 + 4.00Q = 75,000 + 5.00Q$$

$$5.00Q - 4.00Q = 80,000 - 75,000$$

$$1.00Q = 5,000 \implies Q_{BC} = \mathbf{5,000} \text{ units}$$

- At $Q = 5,000$: $TC_B = 80,000 + 4(5,000) = \$100,000$, and $TC_C = 75,000 + 5(5,000) = \$100,000$.

Crossover 2: Process A vs. Process B

Equating $TC_A(Q) = TC_B(Q)$:

$$110,000 + 2.00Q = 80,000 + 4.00Q$$

$$4.00Q - 2.00Q = 110,000 - 80,000$$

$$2.00Q = 30,000 \implies Q_{AB} = \mathbf{15,000} \text{ units}$$

- At $Q = 15,000$: $TC_A = 110,000 + 2(15,000) = \$140,000$, and $TC_B = 80,000 + 4(15,000) = \$140,000$.

Verification: Is Process B ever dominated?

Let's check the direct crossover between Process A and Process C:

$$110,000 + 2.00Q = 75,000 + 5.00Q \implies 3.00Q = 35,000 \implies Q_{AC} \approx 11,667 \text{ units}$$

Notice that $Q_{BC}(5,000) < Q_{AC}(11,667) < Q_{AB}(15,000)$. Because the intersection Q_{AC} lies strictly between Q_{BC} and Q_{AB} , **Process B is NOT dominated** and forms an active middle operating zone.

Final Decision Policy Table

Production Range (Q units/year)	Recommended Process	Economic Justification
$0 \leq Q < 5,000$	Process C	Lowest fixed capital commitment (\$75,000). High variable cost does not accumulate sufficiently to overcome fixed savings.
$Q = 5,000$	Either B or C	Both yield identical total annual cost (\$100,000).
$5,000 < Q < 15,000$	Process B	Optimal balance between fixed setup and moderate variable production cost.
$Q = 15,000$	Either A or B	Both yield identical total annual cost (\$140,000).
$Q > 15,000$	Process A	Heavy initial fixed investment (\$110,000) is rapidly diluted across volume by the ultra-low variable rate (\$2.00/unit).

4. Problem 3: Make-or-Buy Quantitative Decision

Problem Statement

An industrial plant requires 12,000 precision aluminum bushings per year.

- **Option 1 (Buy from Supplier):** Purchase price is \$18.50 per finished bushing, with zero in-house fixed capital costs.
- **Option 2 (Make In-House):** Purchase a dedicated CNC lathe and bar feeder:
 - Annual machine depreciation, floor space & tooling: $FC = \$45,000/\text{year}$.
 - In-house raw bar stock, tooling insert wear, and operator wages: $v = \$14.00/\text{unit}$.

Questions

1. At 12,000 units/year, should the company Make or Buy?
2. What is the Make-or-Buy crossover volume?

Solution

Part 1: Cost Comparison at $Q = 12,000$

$$TC_{\text{Buy}}(12,000) = 18.50 \times 12,000 = \mathbf{\$222,000}$$

$$TC_{\text{Make}}(12,000) = 45,000 + (14.00 \times 12,000) = 45,000 + 168,000 = \mathbf{\$213,000}$$

$$\text{Annual Savings by Making In-House} = \$222,000 - \$213,000 = \mathbf{\$9,000/\text{year}}$$

Decision: Make In-House.

Part 2: Crossover Volume

$$TC_{\text{Buy}}(Q) = TC_{\text{Make}}(Q) \implies 18.50Q = 45,000 + 14.00Q$$

$$4.50Q = 45,000 \implies Q_{\text{crossover}} = 10,000 \text{ units}$$

- If demand $< 10,000$ units/year $\implies$ **Buy.**
- If demand $> 10,000$ units/year $\implies$ **Make.**

5. Problem 4: Multi-Level Bill of Materials (BOM) Explosion

Problem Statement

An industrial furniture factory receives a contract to deliver **250 Industrial Workbenches (Product W)**. The hierarchical BOM structure is defined as follows:

- **Level 0: Workbench W (Qty: 1)**
 - **Tabletop Sub-Assembly T (Qty: 1)**
 - Wood Top (Qty: 1)
 - Steel Edging (Qty: 4)
 - **Leg Frame L (Qty: 2)**
 - Square Tube Sections (Qty: 2 per frame $\implies$ 4 total)
 - Molded Foot Pads (Qty: 2 per frame $\implies$ 4 total)
 - **Hardware Kit H (Qty: 1)**
 - M8 Bolts (Qty: 16)
 - M8 Locking Nuts (Qty: 16)

Required Quantities for 250 End Units

To find total raw parts required, multiply down the tree branches:

Part Code	Description	Quantity per End Unit	Total Required for 250 Workbenches
Sub-Assy T	Tabletop Assembly	1	$250 \times 1 = \mathbf{250}$
Wood Top	Core Timber Surface	$1 \times 1 = 1$	$250 \times 1 = \mathbf{250}$
Steel Edging	Protective Perimeter Trim	$4 \times 1 = 4$	$250 \times 4 = \mathbf{1,000}$
Leg Frame L	Welded Leg Frame	2	$250 \times 2 = \mathbf{500}$
Square Tube	40mm Steel Tube Sections	$2 \times 2 = 4$	$250 \times 4 = \mathbf{1,000}$
Foot Pad	Molded Rubber Leveler	$2 \times 2 = 4$	$250 \times 4 = \mathbf{1,000}$
Kit H	Pre-packaged Hardware	1	$250 \times 1 = \mathbf{250}$
M8 Bolts	High-tensile fasteners	$16 \times 1 = 16$	$250 \times 16 = \mathbf{4,000}$
M8 Nuts	Locking Hex Nuts	$16 \times 1 = 16$	$250 \times 16 = \mathbf{4,000}$

6. Common Quantitative Calculation Pitfalls on Exams

[!WARNING] **Pitfall 1: Confusing Unit Contribution Margin with Total Revenue** In the formula $Q_{BEP} = \frac{FC}{P-v}$, students frequently forget to subtract v from P in the denominator, writing $\frac{FC}{P}$. This is catastrophic and produces a completely wrong volume!

[!WARNING] **Pitfall 2: Forgetting to Check if an Intermediate Process is Dominated** When given 3 processes (like A, B, and C), always check whether the middle process actually has an active zone. If $Q_{BC} > Q_{AB}$, Process B is strictly dominated and should **never** be used at any volume!

[!WARNING] **Pitfall 3: Inverting Fixed Cost Differences** When solving $Q = \frac{FC_1 - FC_2}{v_2 - v_1}$, ensure the numerators and denominators are matched in sign ($FC_{\text{high}} - FC_{\text{low}}$ divided by $v_{\text{high}} - v_{\text{low}}$). Volume must always be a positive integer!